

Exercise 2

Let $(y_1, \dots, y_5)$ and $(y_6, \dots, y_{10})$ be two independent samples from a Poisson distribution of mean $\exp\{\beta_1\}$ and from a Poisson distribution of mean $\exp\{\beta_1 + \beta_2\}$, respectively.

- a) Formulate an appropriate Poisson regression model for the expected value of Y_i , $i = 1, \dots, 10$.
- b) Write the log-likelihood function of $\underline{\beta} = (\beta_1, \beta_2)$ and the score function. Find the maximum likelihood estimate of (β_1, β_2) . Finally, obtain the observed information matrix.
- c) Determine an approximate distribution of the maximum likelihood estimator $\hat{\underline{\beta}}$ of $\underline{\beta} = (\beta_1, \beta_2)$, and an approximate distribution of the maximum likelihood estimator $\hat{\beta}_1$ of β_1 .
- d) Provide the interpretation of the coefficient β_2 .
- e) Define the concept of “saturated model” and obtain the expression of maximum of the log-likelihood for this model.

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- $y_1, \dots, y_5 \quad Y_i \sim \text{Pois}(e^{\beta_1})$
- $y_6, \dots, y_{10} \quad Y_i \sim \text{Pois}(e^{\beta_1 + \beta_2})$

$Y_i \sim \text{POISSON}(\lambda_i)$

$\eta_i = \beta_1 + \beta_2 x_i \longrightarrow \begin{cases} x_i = 0 & \text{for } i = 1, \dots, 5 \\ x_i = 1 & \text{for } i = 6, \dots, 10 \end{cases}$

$\omega(\lambda_i) = \eta_i$

6 LOG-LIKELIHOOD FUNCTION OF $\underline{\beta} = (\beta_1, \beta_2)$

$f(\theta) = \frac{e^{-\lambda_i} \lambda_i^{y_i}}{y_i!} \quad \lambda_i = e^{\beta_1 + \beta_2 x_i} \quad \lambda_i = e^{\tilde{x}_i^T \beta}$

LIKELIHOOD

$L(\underline{\beta}) = \prod \frac{e^{-\lambda_i} \lambda_i^{y_i}}{y_i!}$

LOG-LIKELIHOOD

$l(\underline{\beta}) = \log(L(\underline{\beta})) = \log\left(\prod \frac{e^{-\lambda_i} \lambda_i^{y_i}}{y_i!}\right) = \sum \log\left(\frac{e^{-\lambda_i} \lambda_i^{y_i}}{y_i!}\right) = \sum -\lambda_i + \sum \log(\lambda_i)^{y_i} - \sum \log(y_i!) = \sum -e^{\tilde{x}_i^T \beta} + \sum y_i \tilde{x}_i^T \beta - \sum \log(y_i!)$

SCORE FUNCTION

$l_{\beta}(\underline{\beta}) = \begin{cases} a = \frac{d}{d\beta_1} l(\underline{\beta}) \\ b = \frac{d}{d\beta_2} l(\underline{\beta}) \end{cases} \quad \begin{cases} a = \frac{d}{d\beta_1} l(\underline{\beta}) = \frac{d}{d\beta_1} \left[\sum -e^{\beta_1 + \beta_2 x_i} + \sum y_i (\beta_1 + \beta_2 x_i) - \sum \log(y_i!) \right] = \sum -e^{\beta_1 + \beta_2 x_i} \cdot (1) + \sum y_i \longrightarrow -se^{\beta_1} - se^{\beta_1 + \beta_2} + \sum_{i=1}^{10} y_i \\ b = \frac{d}{d\beta_2} l(\underline{\beta}) = \sum -e^{\beta_1 + \beta_2 x_i} \cdot (x_i) + \sum y_i x_i \longrightarrow 0 - se^{\beta_1 + \beta_2} + \sum_{i=6}^{10} y_i \end{cases}$

$l_{\beta\beta}(\underline{\beta}) = \begin{bmatrix} \frac{d}{d\beta_1} \frac{d}{d\beta_1} l(\underline{\beta}) & \frac{d}{d\beta_1} \frac{d}{d\beta_2} l(\underline{\beta}) \\ \frac{d}{d\beta_2} \frac{d}{d\beta_1} l(\underline{\beta}) & \frac{d}{d\beta_2} \frac{d}{d\beta_2} l(\underline{\beta}) \end{bmatrix} = \begin{bmatrix} -se^{\beta_1} - se^{\beta_1 + \beta_2} & -se^{\beta_1 + \beta_2} \\ -se^{\beta_1 + \beta_2} & -se^{\beta_1 + \beta_2} \end{bmatrix}$

7 DISTRIBUTION OF MLE $\hat{\underline{\beta}}$ OF $\underline{\beta}$

~~$\bullet Y \sim N(\underline{\beta}; \Sigma(\underline{\beta}))$~~

~~$\bullet \hat{\underline{\beta}}(Y) \sim N(\underline{\beta}, [\Sigma(\underline{\beta})]^{-1})$~~

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8 INTERPRETATION OF β_2

d) we already noticed that

for sample 1 $E[Y_i] = \mu_1 = e^{\beta_1}$

for sample 2 $E[Y_i] = \mu_2 = e^{\beta_1 + \beta_2}$

$\frac{\mu_2}{\mu_1} = \frac{e^{\beta_1 + \beta_2}}{e^{\beta_1}} = e^{\beta_2} \Rightarrow \beta_2 = \log \frac{\mu_2}{\mu_1} = \log \frac{E[Y_i | \text{sample 2}]}{E[Y_i | \text{sample 1}]}$

Hence β_2 represents the log of the ratio between the means of the two samples.

Or, noting that $\mu_2 = e^{\beta_2} \cdot \mu_1$

the mean of sample 2 is obtained by multiplying the mean of sample 1 of a coefficient e^{β_2} .

9 SATURATED MODEL

SATURATED MODEL

MODEL WITH 11 COVARIATES

$\tilde{\lambda}(\text{SAT.}) = \sum y_i \log(y_i) - y_i$